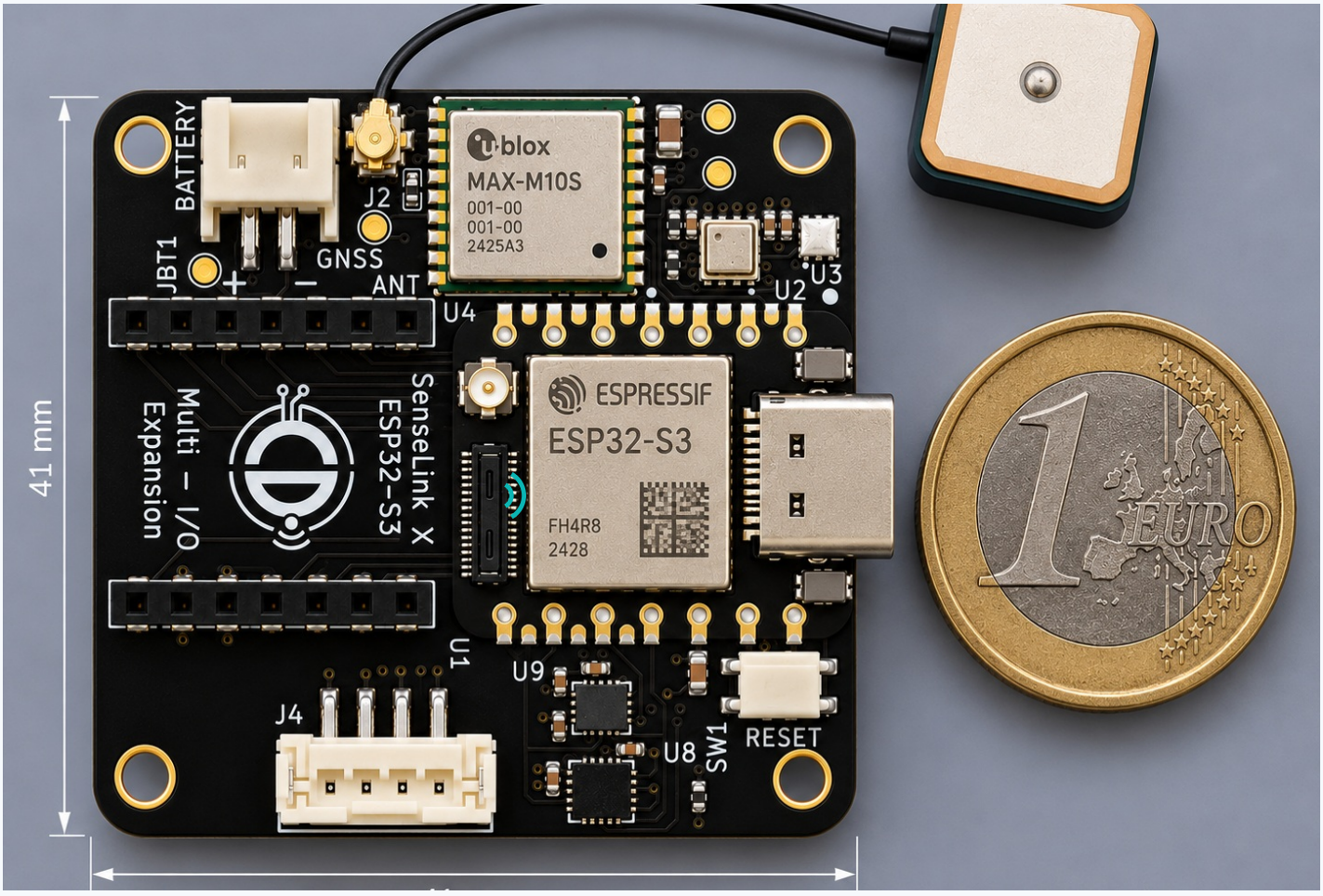
EMBEDDED SYSTEMS PORTFOLIO

ESP32-S3 Wireless Hardware & Embedded Engineering

Hardware Design • Embedded Systems • Wireless • Manufacturing

Engineering of connected embedded hardware from architecture and PCB design through sensor integration, wireless interfaces and production handoff.



GNSS
External active antenna
Positioning / RF

MOTION
BMI323 + ADXL372
6-axis + high-g sensing

LiPo POWER
1-cell battery input
Portable system power

COMPLETE EMBEDDED ENGINEERING CAPABILITY

SCHEMATIC + PCB
Design & implementation

ESP32-S3 SYSTEM
Embedded architecture

WI-FI + BLE
Wireless integration

BOM / CPL / GERBER
Manufacturing handoff

DESIGN VALIDATION
ERC / DRC / review

ESP32-S3 | KiCad | Wi-Fi / BLE | BMI323 | ADXL372 | MAX-M10S | GNSS / RF

Engineering evidence and detailed project work follow.

ESP32-S3 WIRELESS HARDWARE & EMBEDDED ENGINEERING

From System Concept to Manufacturing-Ready Hardware

Architecture, schematic, four-layer PCB, production handoff and embedded integration.

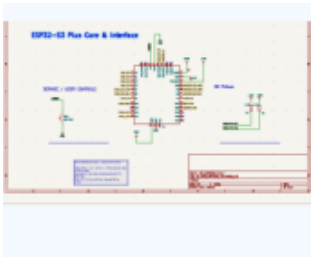
IDEA — ARCHITECTURE — SCHEMATIC — PCB — MANUFACTURING — FIRMWARE — VALIDATION — HARDWARE

1 SYSTEM ARCHITECTURE



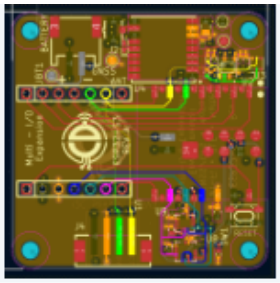
Functional subsystems and interface definitions.

2 SCHEMATIC DESIGN



Power, motion, GNSS and sensor circuits.

3 PCB IMPLEMENTATION



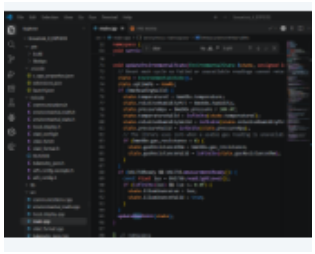
Four-layer placement, routing and access.

4 MANUFACTURING



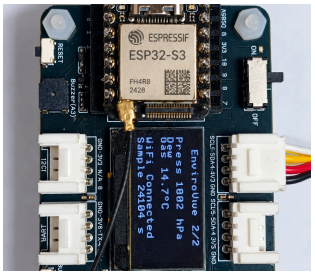
Gerber, BOM and CPL assembly workflow.

5 EMBEDDED FIRMWARE



ESP32-S3 firmware development evidence.

6 DEVELOPMENT BOARD WOR



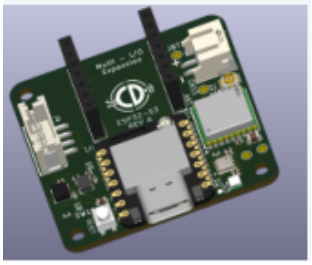
Physical development hardware with live OLED output.

7 APPLICATION INTEGRATION



Connected application interface evidence.

8 COMPACT HARDWARE



41 x 41 mm PCB with GNSS and expansion.

END-TO-END ENGINEERING

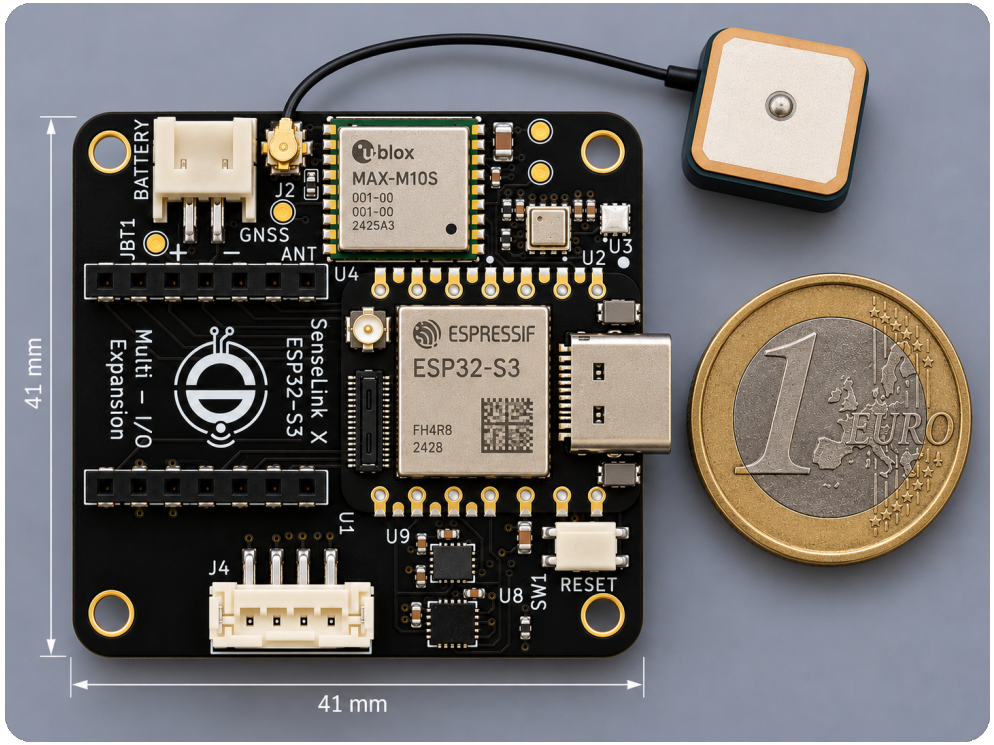
CONCEPT > SCHEMATIC > PCB > MANUFACTURING > FIRMWARE > INTEGRATION

ESP32-S3 | KiCad | Wi-Fi / BLE | GNSS | Sensors | BOM / CPL / Gerbers

CUSTOM ESP32-S3 HARDWARE DESIGN

Wi-Fi / BLE • Motion • Environmental Sensing • GNSS

Four-layer KiCad design integrating wireless connectivity, dual motion sensing, positioning, power and expansion.



ESP32-S3 Rev A | 41 × 41 mm board design

ENGINEERING FEATURE SET

WIRELESS + INTERFACES

ESP32-S3 platform with 2.4 GHz Wi-Fi / BLE, USB and GPIO.

MOTION / CONDITION

BMI323 six-axis IMU and ADXL372 high-g accelerometer.

ENVIRONMENTAL + LIGHT

BME688 environmental and VEML6035 ambient-light sensors.

GNSS + ACTIVE ANTENNA

u-blox MAX-M10S, UART / PPS and external GNSS antenna.

POWER ARCHITECTURE

LiPo battery interface, regulated 3.3 V distribution and USB.

MODULAR EXPANSION

External I²C, SPI, UART and GPIO for add-on peripherals.

KICAD + MANUFACTURING

Hierarchical schematics, four-layer PCB, BOM, CPL and Gerbers.

ESP32-S3 | Wi-Fi + BLE | BMI323 + ADXL372 | BME688 + VEML6035 | MAX-M10S | 4-layer KiCad

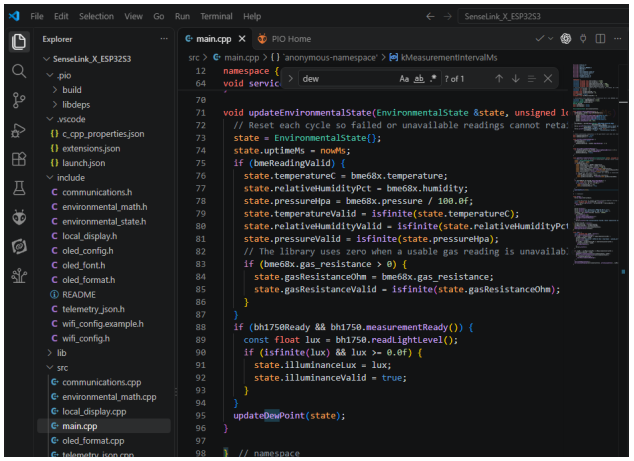
Hardware design and manufacturing evidence; physical validation is documented separately where available.

ESP32-S3 FIRMWARE ENGINEERING

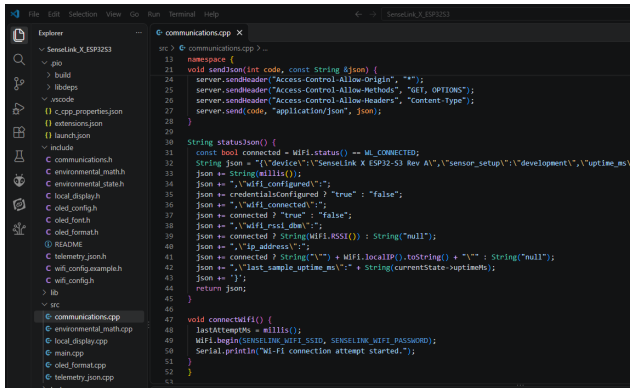
PlatformIO | Embedded C++ | Sensors | Wi-Fi | JSON Telemetry

Modular application firmware demonstrated on physical ESP32-S3 development hardware.

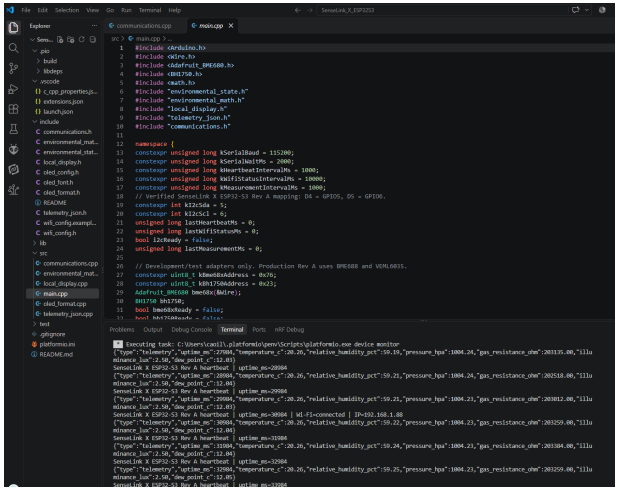
FIRMWARE ARCHITECTURE



SENSOR PROCESSING



LIVE SERIAL TELEMETRY



MODULAR EMBEDDED C++

Separated communications, display, sensor state, environmental maths and JSON telemetry modules.

ENVIRONMENTAL ACQUISITION

BME680 / BH1750 sensor acquisition, data validation and dew-point calculation.

WI-FI + JSON INTERFACE

Wi-Fi connection/status handling, JSON data output, HTTP response structure and status reporting.

HARDWARE TELEMETRY

Serial logs show temperature, humidity, pressure, gas resistance, light and device heartbeat.

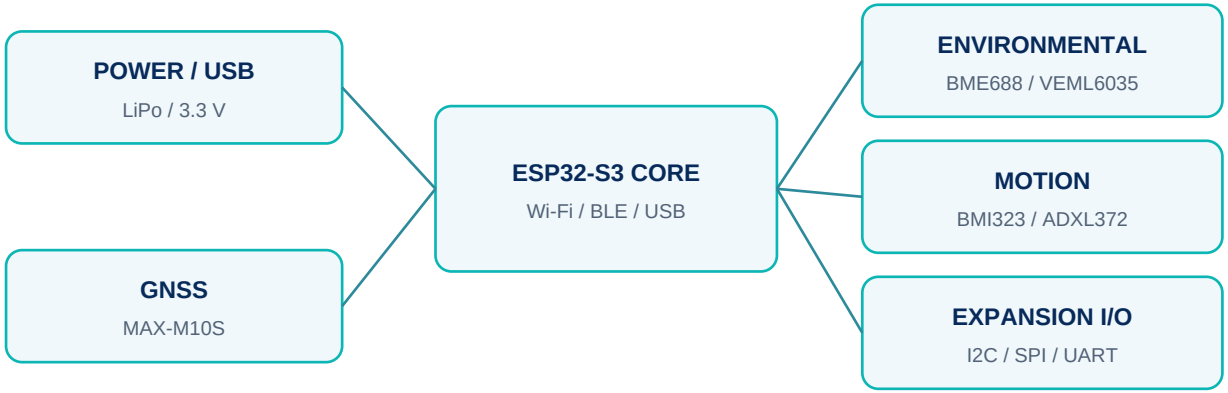
DEVELOPMENT EVIDENCE | PlatformIO / Arduino | I2C | Wi-Fi | OLED | JSON | Serial Debug

SCHEMATIC & ELECTRONIC DESIGN

System Architecture • Power • GNSS / RF

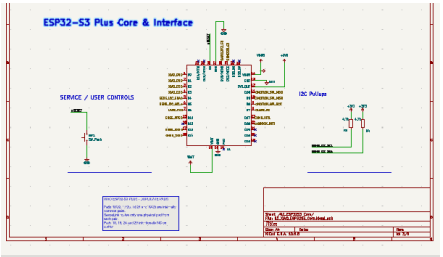
Hierarchical KiCad design organising ESP32-S3 connectivity, motion, environmental sensing, positioning and expansion.

SYSTEM-LEVEL SCHEMATIC ARCHITECTURE

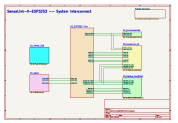


Subsystem partitioning keeps sensor, power and interface boundaries explicit.

ESP32-S3 CORE & INTERFACES



HIERARCHICAL KICAD SCHEMATIC



SYSTEM DECOMPOSITION

Functional blocks separate power, ESP32-S3 core, motion, environment, expansion and GNSS interfaces.

POWER / USB

Battery and USB power entry, regulated system supply and local peripheral distribution.

GNSS / RF

MAX-M10S receiver with external antenna connection and defined UART / timing interfaces.

DESIGN FOR DEBUG

Core service controls, accessible interfaces and structured sheets support bring-up and review.

REQUIREMENTS → SYSTEM ARCHITECTURE → CIRCUIT DESIGN → INTERFACE DEFINITION → PCB IMPLEMENTATION

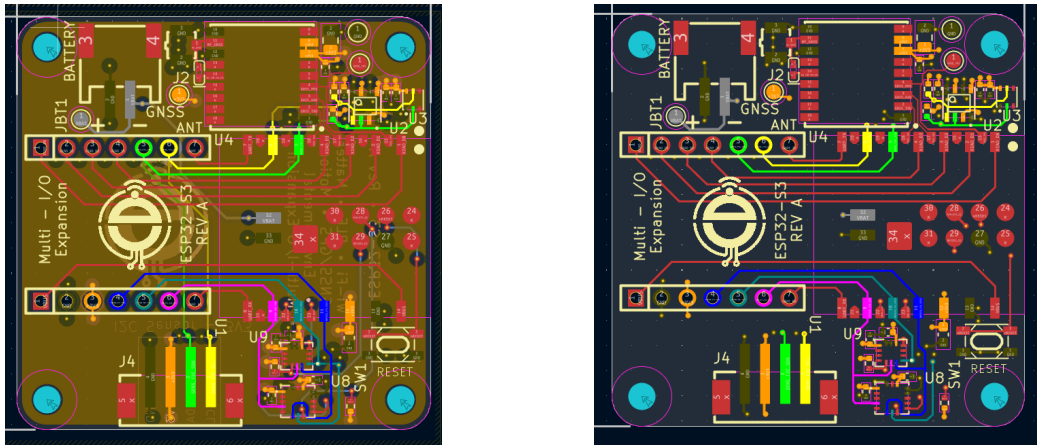
KiCad | ESP32-S3 | I2C / SPI / UART | Motion | GNSS / RF | Expansion

PCB LAYOUT & MANUFACTURING ENGINEERING

Routed PCB • Placement • DFT • Manufacturing Handoff

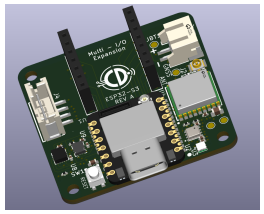
Custom four-layer ESP32-S3 PCB in KiCad, translating system architecture into routed interconnects and fabrication-ready outputs.

ROUTED PCB — COPPER, VIAS & INTERCONNECT



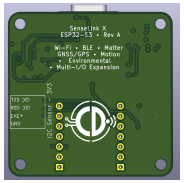
KiCad routed-board and placement views show track distribution, via transitions, sensor / GNSS partitioning and expansion connector access.

3D IMPLEMENTATION



3D assembly review of ESP32-S3 hardware, GNSS module, USB-C, battery input and expansion connectors.

PCB EDITOR / BOARD REVIEW



Top and reverse-side visual checks complement copper routing, connector placement and assembly review.

PLACEMENT & PARTITIONING

The ESP32-S3 core, MAX-M10S GNSS, motion and environmental sensing, battery input and expansion connectors are organized as distinct layout regions.

ROUTING & FAN-OUT

Four-layer implementation exposes signal breakout, copper routing, vias and layer transitions between the MCU, GNSS, sensors and external I/O.

BRING-UP / DFT

USB-C, reset and accessible expansion interfaces provide practical access for firmware loading, electrical checks and bench-level development.

FABRICATION HANDOFF

The design is supported by Gerber/drill exports, a sourced BOM and component-position data for fabrication and assembly review.

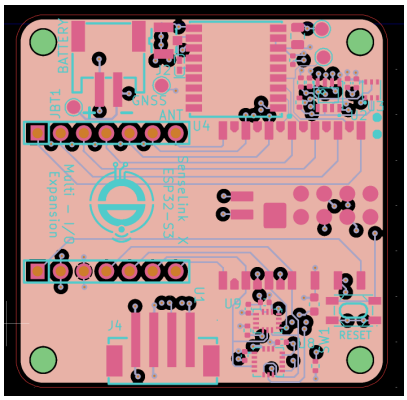
SCHEMATIC → PLACEMENT → ROUTING → DFT / BRING-UP ACCESS → BOM / CPL / GERBERS

KiCad PCB Editor | Four-Layer Layout | ESP32-S3 | GNSS / RF | DFT | Manufacturing Outputs

Assembly-Ready Manufacturing Data

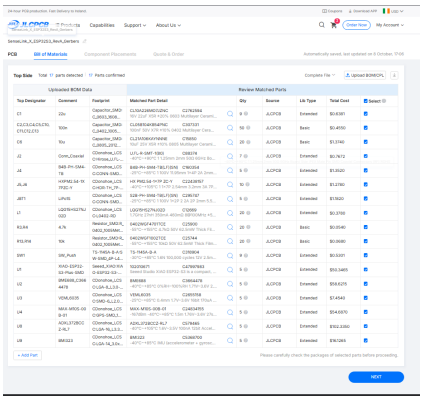
ESP32-S3 Rev A | JLCPCB BOM, CPL, Gerbers & Assembly Review

PCB DESIGN -> GERBER / DRILL -> BOM -> CPL -> JLCPCB ASSEMBLY REVIEW -> RELEASE PACK



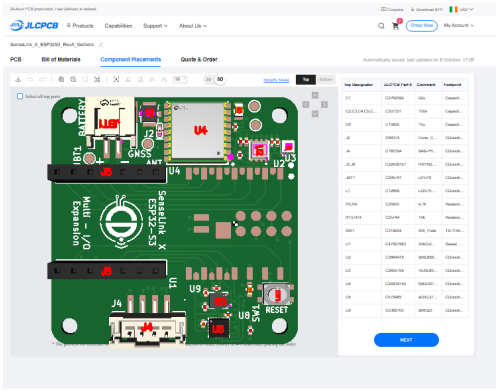
FABRICATION DATA

Gerber Viewer: copper, pads and drill features.



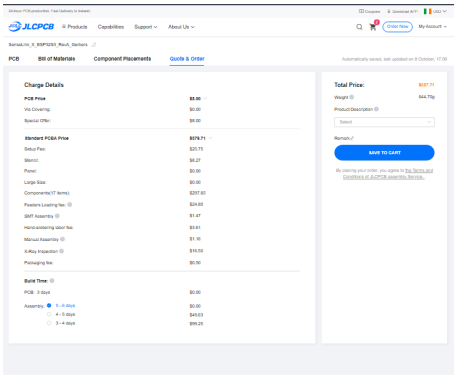
BOM & SOURCING VALIDATION

JLCPCB designator, footprint and parts mapping.



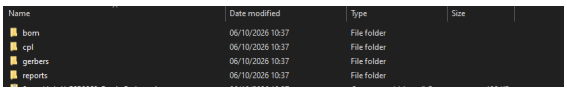
PLACEMENT / CPL REVIEW

JLCPCB pick-and-place and rotation preview.



ASSEMBLY WORKFLOW

JLCPCB PCB + SMT quotation-stage review.



ORGANIZED MANUFACTURING DELIVERABLE

Rev A release separates component sourcing, placement files, Gerber fabrication layers and engineering reports for a traceable handoff.

BOM | CPL | GERBERS | REPORTS

DESIGN -> FABRICATION DATA -> COMPONENT SOURCING -> PLACEMENT -> ASSEMBLY HANDOFF

KiCad | Gerber / Drill | JLCPCB BOM | CPL / Pick & Place | SMT Assembly

WI-FI & CONNECTED TELEMETRY ENGINEERING

ESP32-S3 • Wi-Fi Connectivity • JSON Data • Device Status

Embedded communications code and live serial evidence demonstrating a connected environmental monitoring workflow.

WI-FI COMMUNICATIONS IMPLEMENTATION

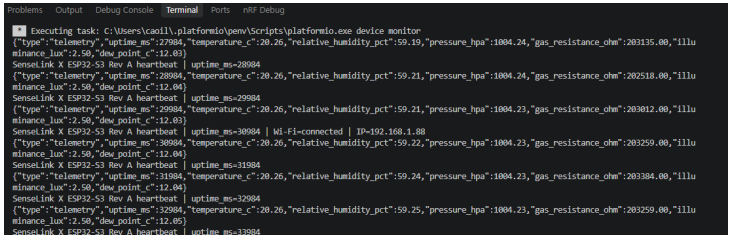
```cpp
void sendJson(int code, const String &json) {
  server.sendHeader("Access-Control-Allow-Origin", "*");
  server.sendHeader("Access-Control-Allow-Methods", "GET, OPTIONS");
  server.sendHeader("Access-Control-Allow-Headers", "Content-Type");
  server.send(code, "application/json", json);
}

String statusJson() {
  const bool connected = WiFi.status() == WL_CONNECTED;
  String json = "{\"device\":\"SenseLink X ESP32-S3 Rev A\",\"sensor_setup\":\"development\",\"uptime_ms\
  json += String(millis());
  json += ",\"wifi_configured\":";
  json += credentialsConfigured ? "true" : "false";
  json += ",\"wifi_connected\":";
  json += connected ? "true" : "false";
  json += ",\"wifi_rssi_dbm\":";
  json += connected ? String(WiFi.RSSI()) : String("null");
  json += ",\"ip_address\":";
  json += connected ? String("\"") + WiFi.localIP().toString() + "\"" : String("null");
  json += ",\"last_sample_uptime_ms\":" + String(currentState->uptimeMs);
  json += '}';
  return json;
}

void connectWifi() {
  lastAttemptMs = millis();
  WiFi.begin(SENSELINK_WIFI_SSID, SENSELINK_WIFI_PASSWORD);
  Serial.println("Wi-Fi connection attempt started.");
}
}
```

Wi-Fi connection handling, status JSON and HTTP response structure in embedded C++.

LIVE DEVICE TELEMETRY



Serial output includes sensor values and Wi-Fi status.

DATA FLOW — SENSOR MEASUREMENT TO CONNECTED INTERFACE

SENSORS — I2C acquisition > **APPLICATION** — Validated state > **JSON** — Structured fields > **WI-FI / HTTP** — Status interface > **CLIENT** — Data consumer

WI-FI STATUS

Connected state, RSSI and IP

JSON TELEMETRY

Environmental measurement fields

MODULAR C++

Separated communications code

LIVE DEBUG

Serial heartbeat and readings

ESP32-S3 | PlatformIO / Arduino | Wi-Fi | Embedded C++ | HTTP / JSON | Serial Telemetry

END-TO-END SYSTEM VALIDATION

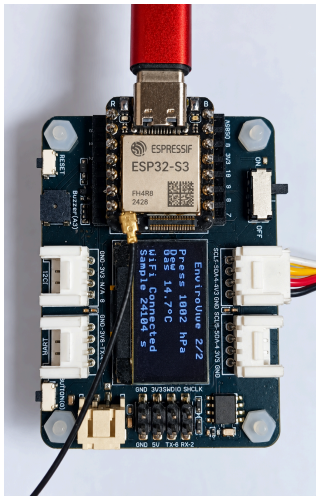
Physical Hardware to Live Application

ESP32-S3 | Sensor Acquisition | Wi-Fi | JSON | Dashboard

Demonstrated environmental measurements across firmware, serial output and a live client interface.

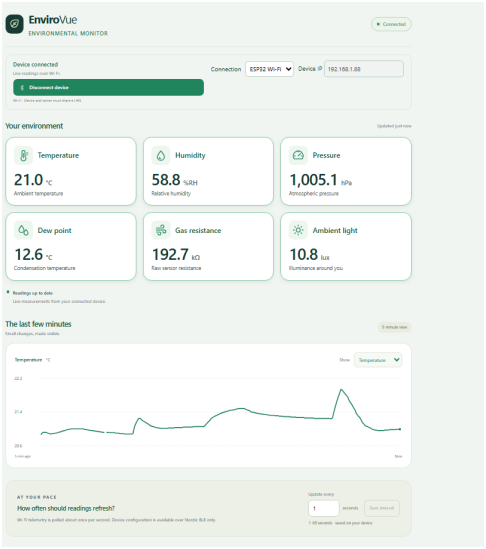
PHYSICAL SENSORS → ESP32-S3 FIRMWARE → JSON / WI-FI → LIVE APPLICATION

PHYSICAL DEVELOPMENT HARDWARE



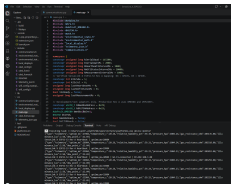
Live OLED values on ESP32-S3 hardware.

ENVIROVUE — LIVE CLIENT



Environmental values and recent trends.

SERIAL TELEMETRY & CONNECTIVITY EVIDENCE



Structured JSON measurements and periodic device heartbeat.

Wi-Fi connection status and IP address reported over serial.

Supports tracing sensor values through the embedded data path.

PHYSICAL INTEGRATION

ESP32-S3 development hardware with connected sensors and local OLED feedback.

LIVE MEASUREMENTS

Temperature, humidity, pressure, gas resistance and light measurements.

CONNECTED DATA PATH

Embedded C++ acquisition, structured JSON, Wi-Fi status and client display.

APPLICATION EVIDENCE

EnviroVue dashboard displays measurements, device connection and trend graph.

SENSOR READINGS → FIRMWARE → CONNECTIVITY → CLIENT VISUALISATION

ESP32-S3 | PlatformIO | Environmental Sensors | Wi-Fi / JSON | EnviroVue

Development-platform validation; production PCB testing is not asserted.